

Abstract. We study top-@Mk gating in mixture-of-experts (MoE) transformers as the context window grows from 8K to 128K tokens. At long contexts the router becomes the bottleneck: load imbalance grows super-linearly with sequence length and routing entropy collapses. We propose *entropy-anchored routing*, a constant-overhead modification that maintains expert utilisation within 8% of uniform across all context lengths we tested, while improving perplexity by 0.34 nats on the PG-19 long-document benchmark.

1. Introduction

Long-context language models stress every subsystem of a transformer, but the routing layer in mixture-of-experts variants is the first to break. At a context of 128K tokens the per-token routing cost is amortised across many more tokens than at 8K, yet the load distribution across experts is observed to grow markedly more uneven. The standard top-@M{k} gate of Shazeer et al. [1] balances load via an auxiliary loss that pulls gating logits toward uniform; at long context this auxiliary loss is dominated by the cross-entropy term and contributes negligibly to gradients.

The contribution of this poster is a single drop-in change to the gate that re-introduces an explicit entropy floor. We call this scheme @I { entropy-anchored routing }. The modification adds 0.4% to the forward-pass FLOPs and no parameters. We evaluate on three open MoE checkpoints (Mixtral-8x7B, Switch-Base-128, and our own 1.3B-A0.3B model) and on four long-context benchmarks.

2. Method

Let $x_t \in \mathbb{R}^d$ be the token representation at position t and let $W_r \in \mathbb{R}^{E \times d}$ be the router weight matrix for E experts. The standard top- k gate computes

$$g_t = \text{softmax}(\frac{1}{\tau} W_r x_t) \propto \sum_e \exp(\frac{1}{\tau} \sum_{i=1}^d W_{r_{ei}} x_{t,i}) / \sum_e \exp(\frac{1}{\tau} \sum_{i=1}^d W_{r_{ei}} x_{t,i})$$

Our modification replaces the softmax with a tempered softmax whose temperature τ_t is chosen, per token, to enforce a target entropy H^* :

$$g_t = \text{softmax}(\frac{1}{\tau_t} W_r x_t) \propto \sum_e \exp(\frac{1}{\tau_t} \sum_{i=1}^d W_{r_{ei}} x_{t,i}) / \sum_e \exp(\frac{1}{\tau_t} \sum_{i=1}^d W_{r_{ei}} x_{t,i})$$

The one-dimensional minimisation is solved to within 0.01 nats in three Newton steps – the entropy is monotone in τ , so convergence is immediate. Figure 1 sketches the routing pipeline.

[inline SVG omitted in non-SVG back-end]

Figure 1. Routing pipeline with the entropy-anchored gate. The tempered softmax block is the only departure from a vanilla top- k gate; downstream experts are untouched.

3. Results

We compare three gating schemes – vanilla top-2, top-2 with load loss [2], and entropy-anchored – on four context lengths. Across all four lengths the entropy-anchored gate maintains expert utilisation within 8% of uniform; the load-loss baseline drifts to 31% imbalance at 128K. Perplexity on PG-19 improves by 0.34 nats over the load-loss baseline at 128K and by 0.07 nats at 8K, suggesting the benefit grows with context.

[inline SVG omitted in non-SVG back-end]

Figure 2. Expert utilisation imbalance as context grows. Vanilla top-2 degrades fastest; the load-loss term [2] only delays the problem. Entropy anchoring is essentially flat.

context	vanilla	load-loss	anchored
8K	4.2%	3.8%	3.7%
16K	7.1%	5.9%	4.1%
32K	13.0%	9.4%	5.3%
64K	22.0%	16.3%	6.8%
128K	35.0%	31.0%	7.9%

4. Discussion

The result that surprised us most was the @I { robustness } of the anchored scheme to choice of H^* : any target between 0.6 and 0.9 times $\log E$ gave indistinguishable downstream perplexity, even though the realised temperatures differed by an order of magnitude. This argues that the auxiliary loss in [1] is solving the right problem – enforce entropy – but with

the wrong tool, a soft penalty whose strength dissolves at long context.

Two threats to validity. First, our 1.3B model is small enough that gate behaviour may not generalise to the 70B+ scale; we plan a replication at 30B. Second, our long-context evaluations rely on PG-19, which is literary text; routing on code or scientific prose may differ.

5. Conclusion

Entropy-anchored routing is a one-line change to the top- k gate that restores the load-balancing property load-loss methods lose at long context. The overhead is negligible (0.4% forward FLOPs, zero parameters) and the perplexity gain grows with context length. Source and pretrained gates are available at the mdlout group GitHub.

References

References

- N. Shazeer et al. Outrageously large neural networks: the sparsely-gated mixture-of-experts layer. *ICLR*, 2017.
- W. Fedus, B. Zoph, N. Shazeer. Switch transformer: scaling to trillion parameter models. *JMLR*, 23(120):1-39, 2022.